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Python Virtual Environments

Project Isolation and Dependency Management

PYTHON PROJECT ENGINEER SERIES

1. What is a Virtual Environment?

A virtual environment is an **isolated container** on your computer where you can run and test your Python projects without variables from one project leaking into another.

The Container Analogy

Imagine each project lives in its own box. Each box has its own version of Python and its own specific set of tools (packages). Tools in Box A do not affect Box B.

- Separate Python interpreters per project.
- Custom sets of installed packages.
- Zero interference with the system Python installation.

2. Why Use Virtual Environments?

Development becomes messy when different projects require different versions of the same library. Virtual environments solve this "dependency hell".

Conflict Prevention: Version 2.0 and 1.0 can coexist on the same machine.

Portability: Easily replicate your environment on other machines using requirement files.

Cleanliness: Keeps your global site-packages folder free of temporary project libraries.

3. Creating an Environment

Python includes a built-in module called `venv` for this purpose. Navigate to your project folder and run:

```
C:\Project> python -m venv myfirstproject
```

The Resulting Structure

Folder/File	Description
Scripts / bin	Contains activation scripts
Lib	Where your installed packages will live
Include	C headers that compile Python packages
pyvenv.cfg	Configuration for the environment

4. Activating the Environment

Creation is not enough; you must tell your terminal to use the isolated interpreter by **activating** it.

Windows (Command Prompt)

```
> myfirstproject\Scripts\activate
```

macOS / Linux

```
$ source myfirstproject/bin/activate
```

Upon activation, your prompt will change to: (myfirstproject) C:\Users>

5. Installing Packages

Once activated, any `pip install` command only affects the virtual environment.

Example: Cowsay

Let's install a fun package that creates ASCII art of a cow:

```
(myfirstproject)> pip install cowsay
```

Notice that the library is downloaded specifically into the `myfirstproject/` `Lib` folder, not your system-wide directories.

6. Using Packages in Venv

Create a file named `test.py` and use the newly installed library.

```
import cowsay

cowsay.cow("Good Mooooorning!")
```

Executing the Script

```
(myfirstproject)> python test.py
```

This will output a stylized ASCII cow directly in your terminal because the code is running inside the environment that "knows" cowsay.

7. Deactivation & Isolation

When you are finished working, use the `deactivate` command to return to your global Python environment.

```
(myfirstproject)> deactivate
```

Proof of Isolation

If you try to run the same script now, it will fail:

```
Traceback (most recent call last):  
  File "test.py", line 1, in <module>  
    import cowsay  
ModuleNotFoundError: No module named 'cowsay'
```

8. Deleting an Environment

Deleting a virtual environment is as simple as deleting its folder. Since it's isolated, no other system settings are affected.

Command Line Deletion (Windows)

```
> rmdir /s /q myfirstproject
```

This completely removes all installed packages and the isolated interpreter associated with that specific project.

9. Practical Exercise

Knowledge Check

Question: What is the built-in Python module used for creating virtual environments?

☐ virt

☐ environ

☒ venv

☐ vrlnv

Module Recap

- Virtual environments prevent version conflicts.
- Activate to use, deactivate to exit.
- Dependencies are contained within the project's folder.

10. Environment Reference Guide

Keep these commands handy for your development workflow.

Task	Command
Create	<code>python -m venv [name]</code>
Activate (Win)	<code>[name]\Scripts\activate</code>
Activate (Unix)	<code>source [name]/bin/activate</code>
Close	<code>deactivate</code>
List Packages	<code>pip list</code>

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